

Linear Equations in One Variable

Mathematics 8 - First Quarter Lecture Notes

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1. Learning Objectives

- Define a linear equation in one variable and identify its standard form.
- Apply the properties of equality to solve equations step by step.
- Translate word problems into linear equations and solve them.

2. Key Definition

A linear equation in one variable is an equation that can be written in the form $ax + b = 0$, where a and b are real numbers and a is not equal to zero. The highest power of the variable is exactly one, which is why its graph would always be a straight line.

Examples: $3x + 5 = 20$, $7 - 2y = 1$, and $5(x - 2) = 3x + 9$ are all linear equations in one variable.

3. Properties of Equality

- Addition Property: adding the same number to both sides preserves the equality.
- Subtraction Property: subtracting the same number from both sides preserves the equality.
- Multiplication Property: multiplying both sides by the same nonzero number preserves the equality.
- Division Property: dividing both sides by the same nonzero number preserves the equality.

4. Steps for Solving

- Step 1: Simplify each side of the equation by removing parentheses and combining like terms.
- Step 2: Move all variable terms to one side using inverse operations.
- Step 3: Move all constant terms to the other side.
- Step 4: Divide both sides by the coefficient of the variable.

- Step 5: Check the answer by substituting it back into the original equation.

5. Worked Examples

Example 1. Solve $3x + 5 = 20$.

- Subtract 5 from both sides: $3x = 15$.
- Divide both sides by 3: $x = 5$.
- Check: $3(5) + 5 = 20$, which is true.

Example 2. Solve $5x - 8 = 2x + 7$.

- Subtract $2x$ from both sides: $3x - 8 = 7$.
- Add 8 to both sides: $3x = 15$.
- Divide both sides by 3: $x = 5$.
- Check: $5(5) - 8 = 17$ and $2(5) + 7 = 17$.

6. Common Mistakes to Avoid

- Forgetting to apply an operation to BOTH sides of the equation.
- Sign errors when moving terms across the equals sign.
- Skipping the checking step, which catches most arithmetic slips.

Next meeting: bring Problem Set 3 (see the Resources tab). We will check Parts A and B together.